ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

MID-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 3

Time Allowed: 1.5 Hours **Maximum Marks: 30**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- PART A ---

Q.1 Explain core OOP concepts: Encapsulation, Abstraction, Inheritance, and Polymorphism. [15 Marks]

[OR]

Q.1 (Alt) Explain Constructors and Destructors: Default, Parameterized, and Copy constructors in C++/Java. [15 Marks]

--- PART B ---

Q.2 Explain Function Overloading vs Function Overriding with virtual functions and dynamic binding. [15 Marks]

[OR]

Q.2 (Alt) Explain Exception Handling mechanisms (try, catch, throw, finally) and Custom Exceptions. [15 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****